Roll No.

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31N0403

B. TECH. III Sem Main/Back Exam 2025-2026

Computer Science and Engineering

(3CS4-03) - Data Structures and Algorithms

3AD4-03, 3AM4-03, 3AI4-03, 3CM4-03, 3CY4-03, 3CD4-03, 3CS4-03, 3IT4-03, 3IC4-03, Common to CS, IT, AI, DS, MC, CM, CD, CA, AD, AM, CY, IO

31N0403

Time : 3 Hours]

[Max. Marks : 70

[Min. Passing Marks :

Instructions to Candidates :

Part-A : Short answer questions (up to 25 words) 10 * 2 marks = 20marks. All ten questions are compulsory.

Part-B : Analytical/Problem Solving questions 5 * 4 marks = 20 marks. Candidates have to answer 5 questions out of 7.

Part-C : Descriptive/Analytical/Problem Solving questions 3 * 10 marks = 30 marks. Candidates have to answer 3 questions out of 5.

Schematic diagrams must be shown wherever necessary. Any data you feel missing may suitably be assumed and stated clearly.

Units of quantities used/calculated must be stated clearly.

Use of following supporting materials is permitted during examination.

(Mentioned in form No. 205)

1 NIL

2 NIL

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Part-A

10x2=20

Q.1 Define time complexity. [2]

Q.2 What are asymptotic notations? [2]

Q.3 Differentiate between stack and queue. [2]

Q.4 State two applications of linked lists. [2]

Q.5 What is the difference between internal and external sorting? [2]

Q.6 Define stable and unstable sorting with one example each. [2]

Q.7 Define the term "height" of a tree. [2]

Q.8 Write any two properties of a binary search tree (BST). [2]

Q.9 What is hashing. [2]

Q.10 What is AVL tree? [2]

Part-B

5x4=20

Q.1 Write an algorithm for Bubble Sort in array. [4]

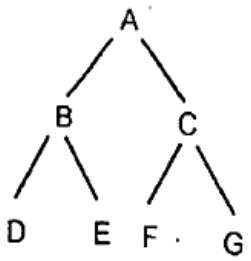
Q.2 Write an alogrithm for PUSH and POP in Stack. [4]

Q.3 Evaluate the following postfix expression using stack- [4]

8,2,+,3,1,-,*,2,2,+,/

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Q.4 Perform inorder, preorder and postorder traversals on the following binary tree: [4]



Q.5 Explain the concept of AVL tree rotation with a suitable example. [4]

Q.6 Write the difference between Single linked and Double linked List. [4]

Q.7 What are the collision resolution techniques in hashing. [4]

Part-C

3x10=30

Q.1 Write an algorithm to insert and delete element in Queue. [10]

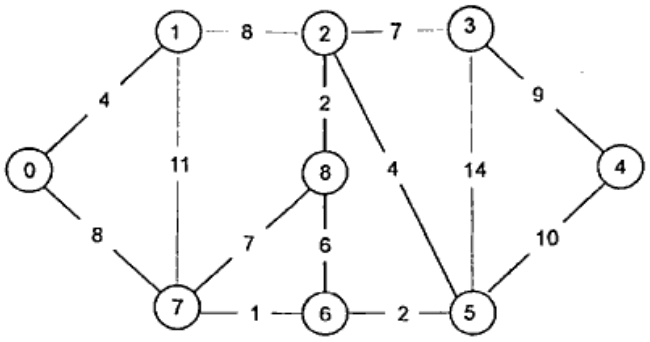
Q.2 Perform Merge sort on following elements 38,27,43,3,9,82,10,14,7,56,21. [10]

Q.3 Write and explain the algorithm for insertion and deletion of node at start of the linked list. [10]

Q.4 Write the BFS algorithm for traversal in the graph. [10]

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Q.5 Explain Prim's algorithm for finding the minimum spanning tree in the given graph. [10]



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